

Case Study: Smart Waste Management System in Urban Areas

1. Introduction

Rapid urbanization has led to inefficient waste management systems, causing pollution and health hazards.

2. Problem Identification

Waste collection is inefficient due to fixed schedules, lack of real-time monitoring, and poor segregation.

3. Root Cause Analysis

Using the 5 Whys technique, the root issue is lack of smart monitoring and low technological adoption.

4. Creative Solutions

1. Smart Dustbins (IoT) 2. AI Route Optimization 3. Reward-Based App 4. Community Composting 5. Drone Monitoring

5. Evaluation (FVD Framework)

Solution	Feasibility	Viability	Desirability
Smart Dustbins	High	Medium	High
AI Routing	High	High	Medium
Reward App	Medium	High	High
Composting	High	High	Medium
Drone Monitoring	Low	Medium	Low

6. Conclusion

Combining IoT and citizen engagement provides an effective and scalable waste management solution.